Rakesh Naik

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Research Summary

Experimental photonics researcher specializing in plasmon-enhanced spectroscopy and machine learning-integrated optical diagnostics. My work focuses on designing nano-engineered SERS substrates using glancing angle deposition (GLAD) and developing data-driven spectral classification frameworks for rapid respiratory virus detection. I integrate nanofabrication, Raman spectroscopy, electromagnetic simulations (COMSOL), and supervised learning pipelines to build clinically translatable diagnostic platforms. My long-term research vision lies at the intersection of plasmonics, nanophotonics, and intelligent optical sensing systems for biomedical and environmental applications.

Education

PhD (Ongoing) — Aug 2024 – Present
Centre for Photonics and Quantum Communication Technology, IIT Roorkee, India

BS-MS (Dual Degree) in Physics — 2017 – 2022
IISER Berhampur, India
CGPA: 9.25 / 10 (Distinction)

Research Experience

Machine Learning-Enabled SERS Platform for Respiratory Virus Detection
Supervisor: Dr. Sachin Kumar Srivastava — 2023 – Present

- Fabrication of nano-sculptured thin films (nSTFs) via GLAD-based thermal evaporation for plasmonic enhancement.
- Developed standardized SERS spectral database for Influenza A, SARS-CoV-2 (Delta/Omicron), RSV, and Rhinovirus.
- Designed ML pipelines for spectral preprocessing, feature extraction, classification, and cross-validation.
- Integrated automated GUI-based diagnostic system with report generation.
- Performed electromagnetic simulations using COMSOL Multiphysics for near-field optimization.
- Contributing to clinical validation against RT-PCR confirmed samples.

Master's Thesis: Detector Simulations for Electron-Ion Collider (EIC) — 2021 – 2022

- Simulated $e + p$ collisions using PYTHIA and conducted detector-level analysis.
- Performed feasibility studies for proposed Time-of-Flight detector systems.
- Analyzed mid-rapidity particle spectra relevant to baryon transport physics.

International Academic Exposure

Sakura Science Exchange Program (Japan) — 2025

- Selected participant in the Sakura Science Program promoting Indo-Japan scientific collaboration.
- Engaged in laboratory visits, research discussions, and academic exchange in advanced photonics and nanotechnology.
- Explored collaborative research opportunities in plasmonics and optical sensing technologies.

Technical Expertise

Nanofabrication: GLAD thin films, plasmonic nano-structures
Optical Techniques: Raman Spectroscopy, SERS, spectral data analysis
Simulation: COMSOL Multiphysics, Geant4, ROOT, PYTHIA 6/8
Programming: Python, C/C++, MATLAB
Machine Learning: Scikit-learn, supervised classification, spectral preprocessing
Scientific Tools: LaTeX, Autodesk Fusion 360

Research Interests

Plasmonics | Nanophotonics | Intelligent Optical Sensing
Machine Learning for Spectroscopy | Biosensing Platforms
Translational Photonic Diagnostics

Selected Academic Activities

ALICE-India School on Quark-Gluon Plasma (2021)
Software Tools in Experimental High Energy Physics (2021)
International Conference on Advances in Functional Materials (2020)

Publications

Journal Articles

1. Arti Yadav, **Rakesh Naik**, Ekta Gupta, Partha Pratim Roy, Sachin Kumar Srivastava, "Single-shot, Receptor-Free, Rapid Detection and Classification of Five Respiratory Viruses by Machine Learning Integrated SERS Sensing Platform," *Biosensors and Bioelectronics*, vol. 279, p.117394, 2025. DOI: 10.1016/j.bios.2025.117394

Book Chapters

1. Arti Yadav†, **Rakesh Naik**†, Sachin Kumar Srivastava, "Artificial Intelligence in Developing Novel SERS-Based Pathogen Sensing," in *Advanced Photonic Devices for Energy and Sensing Applications*, 1st Edition, CRC Press, 2026. † Equal contribution

Patents

1. Sachin Kumar Srivastava, Arti Yadav, **Rakesh Naik**, Ekta Gupta, Partha Pratim Roy, "Single-shot, Rapid Detection and Classification of Influenza, SARS-CoV-2 (Omicron and Delta Variants), RSV and Rhino Viruses in Nasopharyngeal Swab Samples Using Machine Learning-enabled SERS Sensing," Indian Patent Application No. 202411045160, Publication Date: 22 November 2024.

Conference Publications / Presentations

1. **Rakesh Naik**, Arti Yadav, Ekta Gupta, Partha Pratim Roy, Sachin Kumar Srivastava, "Explainable AI-based SERS Sensor for Multi-virus Detection," Workshop on Computational Techniques in Optics & Photonics and Workshop on Sensors & Nanophotonics (OPTCT + SeNcity), IIT Roorkee, India, 2026.
2. Arti Yadav, **Rakesh Naik**, Sagar Kumar Verma, Ekta Gupta, Partha Pratim Roy, Sachin Kumar Srivastava, "Addressing Complexity and Variability Issues of SERS Spectra of Clinical Nasopharyngeal Swab (CNS) Samples for Respiratory Viruses Detection Using Machine Learning," Conference on Lasers and Electro-Optics (CLEO 2025), California, USA.
3. Arti Yadav, **Rakesh Naik**, Ekta Gupta, Partha Pratim Roy, Sachin Kumar Srivastava, "Machine Learning-Integrated SERS System for Rapid Classification of SARS-CoV-2 Variants in Clinical Nasopharyngeal Swab (CNS) Samples," Optica Sensing Congress (OSC 2025), California, USA.
4. Arti Yadav, **Rakesh Naik**, Ekta Gupta, Partha Pratim Roy,Sachin Kumar Srivastava "Machine Learning-integrated SERS Platform for Single-Shot, Receptor-Free Rapid Detection and Classification of Respiratory Viruses," Research Scholar Day, Department of Physics, 2025.
5. Arti Yadav, **Rakesh Naik**, Ekta Gupta, Partha Pratim Roy, Sachin Kumar Srivastava, "Machine Learning-augmented SERS Platform for Single-Shot, Receptor-Free Rapid Detection and Classification of Respiratory Viruses," 1st Annual Virology Summit, ILBS, Delhi, India, 8–9 October 2024.
6. **Rakesh Naik**, Sachin Kumar Srivastava "Machine Learning-Based Detection and Classification of Bacterial and Fungal Species and Their Mixture through Raman Spectra Analysis," Workshop on Computational Techniques in Optics & Photonics and Workshop on Sensors & Nanophotonics (OPTCT + SeNcity), IIT Roorkee, India, 2024.